

Scale-Optimized Geometric Acceleration of Polygonal Approximations to π

C. Wayne Baker
Independent Researcher

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Abstract

Let

$$P_N = N \sin\left(\frac{\pi}{N}\right)$$

be the half-perimeter of the regular $2N$ -gon inscribed in the unit circle. Its error contains only even powers of N^{-1} . For an integer scale factor $b \geq 2$ and an order $r \geq 1$, we form

$$\hat{P}_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} P_{b^m N},$$

where the weights are the Lagrange extrapolation weights at $x = 0$ for the geometric nodes $x_m = b^{-2m}$. The first $r - 1$ even-power errors cancel, so

$$\hat{P}_{N,b}^{[r]} - \pi = O(N^{-2r}).$$

For geometric nodes, the weights are uniformly stable in r :

$$\sum_{m=0}^{r-1} |w_m^{(r,b)}| \leq S_b, \quad S_b = \prod_{\ell=1}^{\infty} \frac{1 + b^{-2\ell}}{1 - b^{-2\ell}}.$$

An explicit sine-remainder bound then quantifies the accelerated error. High-precision computations verify the predicted slopes through $r = 24$ for $b = 2, 3, 6, 9, 27$.

The constructibility status is separated carefully. The dyadic branch is exactly compass-and-straightedge constructible by nested square roots. Exact tripling requires a cubic inversion and is not generally constructible. A constructible linear chord seed has error $O(N^{-3})$, and a general perturbation theorem states the accuracy required of any approximate tripling mechanism to preserve an r -scale hierarchy. No exact construction of every regular $3N$ -gon is claimed.

1 Introduction and Claim Boundary

The classical polygon sequence

$$P_N = N \sin\left(\frac{\pi}{N}\right)$$

converges to π with second-order error. The important fact is not only that the error is small, but that it is structured:

$$P_N = \pi + \frac{C_2}{N^2} + \frac{C_4}{N^4} + \frac{C_6}{N^6} + \cdots.$$

This permits exact cross-scale cancellation of successive error terms.

The paper has four principal claims.

- (i) Geometric scale weights cancel the first $r - 1$ even-power modes.
- (ii) For nodes b^{-2m} , the total weight amplification is uniformly bounded in r .
- (iii) The remaining analytic error has an explicit factorial bound.
- (iv) Scale selection is a tradeoff: increasing b improves weight stability but increases geometric work.

The word *geometric* refers to the polygonal source sequence and its scale hierarchy. It does not imply that every scale branch is exactly straightedge-and-compass constructible. The branch $b = 2$ is exact and constructible. Branches containing a factor of 3 are analytic or numerical scale architectures unless a separate approximate trisection construction is supplied with a sufficient error certificate.

2 The Even-Power Polygonal Expansion

Using the sine series,

$$\sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots,$$

with $z = \pi/N$, one obtains

$$P_N = \pi - \frac{\pi^3}{6N^2} + \frac{\pi^5}{120N^4} - \frac{\pi^7}{5040N^6} + \cdots.$$

More generally,

$$P_N = \pi + \sum_{j=1}^{\infty} (-1)^j \frac{\pi^{2j+1}}{(2j+1)!} N^{-2j}.$$

Only even powers of N^{-1} occur because $\sin z/z$ is an even analytic function.

Definition 2.1 (Even-power error structure). *A sequence $Q_N \rightarrow Q$ has an even-power error structure if*

$$Q_N = Q + C_2 N^{-2} + C_4 N^{-4} + C_6 N^{-6} + \cdots.$$

3 Scale- b Acceleration

Fix an integer $b \geq 2$ and an order $r \geq 1$. Define

$$x_m = b^{-2m}, \quad m = 0, 1, \dots, r-1.$$

The Lagrange weights for evaluating at $x = 0$ are

$$w_m^{(r,b)} = \prod_{\substack{0 \leq i \leq r-1 \\ i \neq m}} \frac{-x_i}{x_m - x_i}.$$

The accelerated polygonal approximation is

$$\hat{P}_{N,b}^{[r]} = \sum_{m=0}^{r-1} w_m^{(r,b)} P_{b^m N}.$$

The weights satisfy

$$\sum_{m=0}^{r-1} w_m^{(r,b)} = 1$$

and, for $j = 1, \dots, r-1$,

$$\sum_{m=0}^{r-1} w_m^{(r,b)} b^{-2jm} = 0.$$

Theorem 3.1 (Scale- b cancellation). *Suppose*

$$Q_N = Q + \sum_{j=1}^{r-1} C_{2j} N^{-2j} + O(N^{-2r}).$$

Then

$$\hat{Q}_{N,b}^{[r]} := \sum_{m=0}^{r-1} w_m^{(r,b)} Q_{b^m N} = Q + O(N^{-2r}).$$

Proof. Substitution gives

$$\hat{Q}_{N,b}^{[r]} = Q \sum_m w_m^{(r,b)} + \sum_{j=1}^{r-1} C_{2j} N^{-2j} \sum_m w_m^{(r,b)} b^{-2jm} + O(N^{-2r}).$$

The unity condition preserves Q , and the moment conditions cancel the first $r-1$ error modes. $\square$

For $r = 2$,

$$\hat{P}_{N,b}^{[2]} = P_{bN} + \frac{P_{bN} - P_N}{b^2 - 1} = \pi + O(N^{-4}).$$

For $b = 2$ and $r = 3$,

$$\hat{P}_{N,2}^{[3]} = \frac{1}{45} P_N - \frac{4}{9} P_{2N} + \frac{64}{45} P_{4N} = \pi + O(N^{-6}).$$

The coefficient $1/45$ is essential; a surviving draft incorrectly displayed $1/64$.

4 Geometric-Node Weight Stability

Set $q = b^{-2}$ and use the notation

$$(q; q)_n = \prod_{\ell=1}^n (1 - q^\ell), \quad (q; q)_0 = 1.$$

Lemma 4.1 (Closed weight formula). *Let $s = r - 1 - m$. Then*

$$|w_m^{(r,b)}| = \frac{q^{s(s+1)/2}}{(q; q)_m (q; q)_s}.$$

Proof. Split the defining product into factors with $i < m$ and $i > m$. After factoring the powers of q , the remaining products are precisely $(q; q)_m$ and $(q; q)_s$. $\square$

Theorem 4.2 (Uniform stability). *For every $r \geq 1$,*

$$\sum_{m=0}^{r-1} |w_m^{(r,b)}| \leq S_b,$$

where

$$S_b = \prod_{\ell=1}^{\infty} \frac{1 + b^{-2\ell}}{1 - b^{-2\ell}} < \infty.$$

Proof. Writing $s = r - 1 - m$ and using $(q; q)_{r-1-s} \geq (q; q)_\infty$,

$$\sum_{m=0}^{r-1} |w_m^{(r,b)}| \leq \frac{1}{(q; q)_\infty} \sum_{s=0}^{\infty} \frac{q^{s(s+1)/2}}{(q; q)_s}.$$

Euler's product identity gives

$$\sum_{s=0}^{\infty} \frac{q^{s(s+1)/2}}{(q; q)_s} = (-q; q)_\infty.$$

Therefore the bound is

$$\frac{(-q; q)_\infty}{(q; q)_\infty} = \prod_{\ell=1}^{\infty} \frac{1 + q^\ell}{1 - q^\ell}.$$

□

The corrected constants for the tested branches are shown in Table 1.

b	$q = b^{-2}$	S_b
2	1/4	1.969260353668269...
3	1/9	1.285210586391831...
6	1/36	1.058822194456829...
9	1/81	1.025316406852818...
27	1/729	1.002751031629743...

Table 1: Uniform stability constants. The May concept PDF rounded the $b = 2$ constant incorrectly; the value above is independently recomputed.

5 Explicit Analytic Remainder Bound

Theorem 5.1 (Sine-remainder bound). *Let $N \geq 6$, $b \geq 2$, and $r \geq 1$. Then*

$$\left| \hat{P}_{N,b}^{[r]} - \pi \right| \leq \frac{\pi^{2r+1}}{(2r+1)!} N^{-2r} \sum_{m=0}^{r-1} |w_m^{(r,b)}| b^{-2rm}.$$

Consequently,

$$\left| \hat{P}_{N,b}^{[r]} - \pi \right| \leq S_b \frac{\pi^{2r+1}}{(2r+1)!} N^{-2r}.$$

Proof. At scale $b^m N$, subtract the first $r - 1$ nonconstant sine terms. Since $\pi/(b^m N) \leq \pi/6 < 1$, the alternating Taylor remainder is bounded by its first omitted term:

$$|R_r(b^m N)| \leq \frac{\pi^{2r+1}}{(2r+1)!} (b^m N)^{-2r}.$$

The acceleration weights cancel the retained terms. Applying the triangle inequality yields the first estimate; dropping $b^{-2rm} \leq 1$ and invoking the stability theorem gives the second. □

6 High-Precision Verification

The accompanying script evaluates the weights, moment residuals, stability constants, accelerated errors, and observed slopes. For a fixed b and r , the slope is obtained by a least-squares fit of

$$\log_{10} |\hat{P}_{N,b}^{[r]} - \pi|$$

against $\log_{10} N$ over $N = 16, 20, 24, 32$. The predicted slope is $-2r$.

Scale b	Order r	Observed slope
2	24	$-47.9999784973 \dots$
3	24	$-47.9999818571 \dots$
6	24	$-47.9999834123 \dots$
9	24	$-47.9999836714 \dots$
27	24	$-47.9999838509 \dots$

Table 2: Representative 1300-digit slope audit. The predicted value is -48 .

These computations are checks of the formulas, not substitutes for the proofs. Their main use is to confirm that high-order cancellation remains numerically visible when sufficient working precision is used.

7 Scale Choice: Stability Versus Work

The stability constant satisfies

$$S_b \longrightarrow 1 \quad (b \rightarrow \infty),$$

so larger scale ratios reduce coefficient amplification. This does not by itself identify an optimal scale. If the cost of constructing or evaluating a polygon is proportional to its side count, a simple work proxy is

$$W_b(r) = \sum_{m=0}^{r-1} b^m = \frac{b^r - 1}{b - 1}.$$

For fixed r , $W_b(r)$ grows rapidly with b , while S_b improves only toward 1. Thus scale selection balances at least three quantities:

scale separation, weight stability, construction cost.

There is no universal optimum without a specified cost model. Earlier claims that one tested branch is globally “best” should be read only relative to the particular empirical normalization used in that computation.

8 Constructibility Status

Let

$$c_N = 2 \sin\left(\frac{\pi}{N}\right), \quad P_N = \frac{N}{2} c_N.$$

For the dyadic branch, starting from $c_6 = 1$, the stable bisection recurrence is

$$c_{2N} = \frac{c_N}{\sqrt{2 + \sqrt{4 - c_N^2}}}.$$

It uses only rational operations and square roots. Therefore all dyadic polygon values and all rationally weighted dyadic accelerants are compass-and-straightedge constructible.

For exact tripling, the triple-angle identity gives

$$c_N = 3c_{3N} - c_{3N}^3.$$

Hence $u = c_{3N}$ is the small positive root of

$$u^3 - 3u + c_N = 0.$$

This cubic is not generally solvable by compass and straightedge. In particular, the regular 18-gon is not constructible, so

$$6 \rightarrow 18 \rightarrow 54 \rightarrow \dots$$

cannot be presented as a ladder of exact constructible regular polygons.

8.1 A Constructible Linear Seed

Proposition 8.1 (Linear tripling seed). *Define*

$$\tilde{c}_{3N}^{(0)} = \frac{c_N}{3}.$$

This segment is constructible from c_N , and

$$\tilde{c}_{3N}^{(0)} - c_{3N} = -\frac{8\pi^3}{81}N^{-3} + O(N^{-5}).$$

In particular, the chord seed error is $O(N^{-3})$.

Proof. Set $x = \pi/N$. Then

$$\frac{c_N}{3} = \frac{2}{3} \sin x = \frac{2x}{3} - \frac{x^3}{9} + O(x^5),$$

whereas

$$c_{3N} = 2 \sin(x/3) = \frac{2x}{3} - \frac{x^3}{81} + O(x^5).$$

Subtract and replace x by π/N . □

8.2 Perturbation Requirement

Suppose the chord used at scale $M = b^m N$ is perturbed by δ_M . Since $P_M = (M/2)c_M$, the accelerated perturbation is bounded by

$$\frac{N}{2} \sum_{m=0}^{r-1} |w_m^{(r,b)}| b^m |\delta_{b^m N}|.$$

Theorem 8.2 (Sufficient perturbation exponent). *For fixed b and r , suppose*

$$|\delta_M| \leq AM^{-\alpha}$$

for all scales used by the operator. If

$$\alpha \geq 2r + 1,$$

then the perturbation contribution is $O(N^{-2r})$ and therefore does not change the formal order of the accelerated hierarchy.

Proof. Substitution gives

$$\frac{N}{2} \sum_m |w_m| b^m |\delta_{b^m N}| \leq \frac{A}{2} N^{1-\alpha} \sum_m |w_m| b^{m(1-\alpha)}.$$

For fixed r , the finite weighted sum is independent of N . If $\alpha \geq 2r + 1$, then $N^{1-\alpha} = O(N^{-2r})$. $\square$

Remark 8.3 (Conditional cubic refinement). If an independently documented constructible refinement map has a local error law

$$e_+ = Ce^3 + O(e^5)$$

and if the $O(N^{-3})$ seed lies in its uniform local basin, then after s refinements its error exponent becomes 3^{s+1} . The exponent condition

$$3^{s+1} \geq 2r + 1$$

is sufficient asymptotically. This statement is conditional on the actual geometric realization and its domain; the present paper does not treat odd symmetry alone as proof that the linear term vanishes, nor does it claim a standalone exact trisection operator.

9 Limitations

The fixed- r acceleration law is polynomial in the resolution variable and is not a competitor to AGM or modular-form algorithms as an arithmetic method for computing π . The numerical branch comparison depends on the selected cost normalization. Exact tripling data generated by cubic root-solving are valid for testing the scale architecture but are not straightedge-and-compass constructions. A fully geometric tripling implementation requires a separate, explicit construction and an error certificate satisfying the perturbation condition above.

The paper also does not claim that the cancellation algebra is new. It is a Richardson-type extrapolation. The contribution is the geometric-node stability analysis, the explicit remainder control, the scale/work comparison, and the precise separation of exact dyadic constructibility from approximate tripling.

10 Conclusion

The polygonal sequence is a scale-indexed carrier of removable error structure. The scale- b operator cancels the first $r - 1$ even-power modes, its geometric-node weights are uniformly stable, and the remaining sine tail has an explicit factorial bound. Larger scales improve weight stability but also increase work, so optimization requires a cost model rather than a largest-scale rule.

The constructibility conclusion is equally precise. Dyadic refinement is exactly constructible. Exact tripling is generally not. Approximate tripling can nevertheless be studied quantitatively: the linear chord seed is constructible with $O(N^{-3})$ error, and any refinement mechanism meeting the required perturbation exponent preserves the accelerated order.

The circle is approached not only by refinement, but by cancellation of refinement error.

References

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